

# Loan Default Risk — Portfolio Health

857M

Total Originated

19.96%

Default Rate

0.73

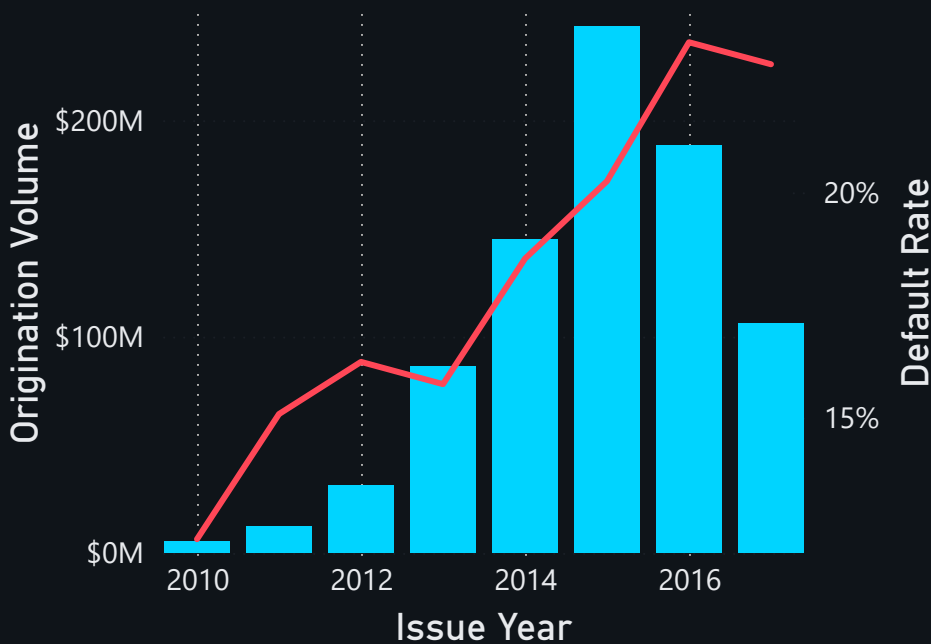
Model AUC

16x

Risk Separation

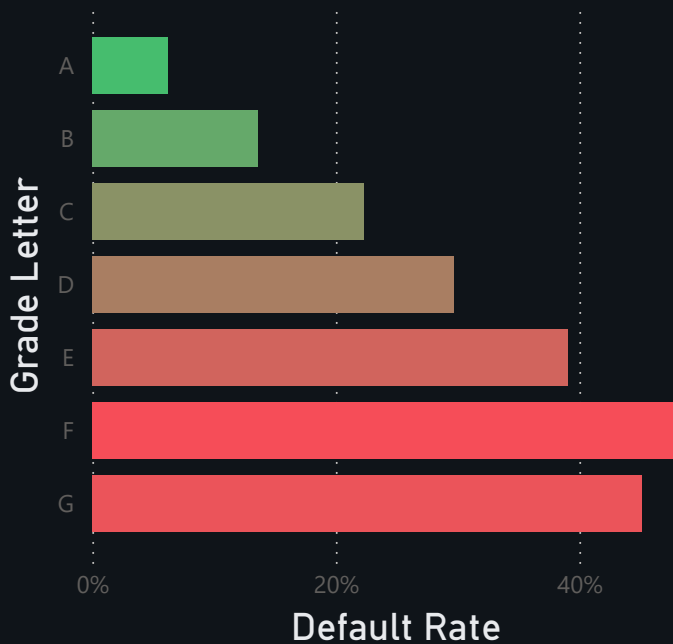
Origination volume vs default rate by vintage

● Sum of loan\_amnt ● Average of actual\_default

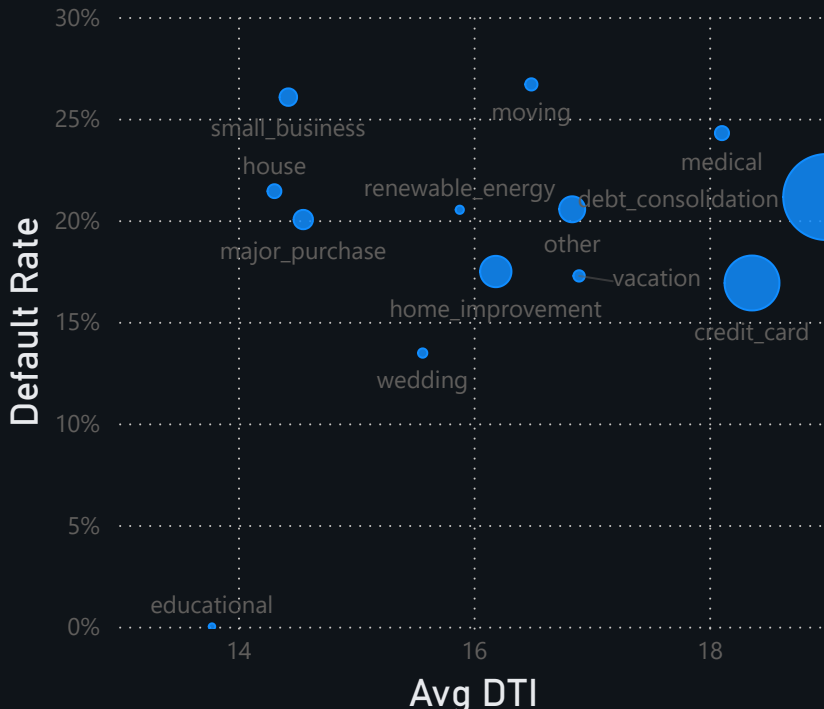


Default rate by grade

Average of actual\_default 0.5% 50.0%



Purpose risk profile



# Risk Decile Lab — What if you priced this differently?

Test-set evaluation · 59,460 loans · \$857M originated · 2007–2018

Approval Threshold

0.00

0.99

LGD Slider

0.50

0.90



829M

Approved Volume

27M

Rejected Volume

0.19

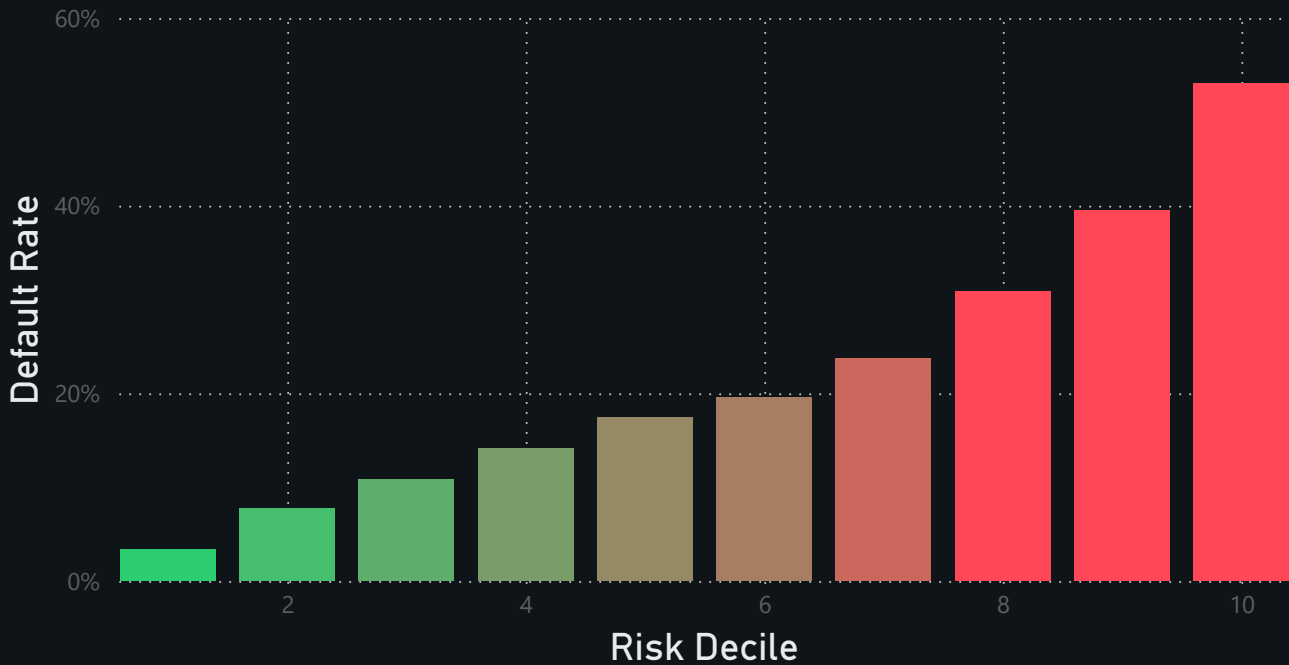
Approved Default Rate

296.19M

Net Margin

Default rate by risk decile

Default Rate Actual 5.00% 30.00%



Cumulative defaults captured by top deciles



# Segment Mispricing — Where the grade system fails

Sub-segments where actual default rate diverges from grade-implied risk

What we charge vs what defaults — by grade



## Default rate by grade \* purpose

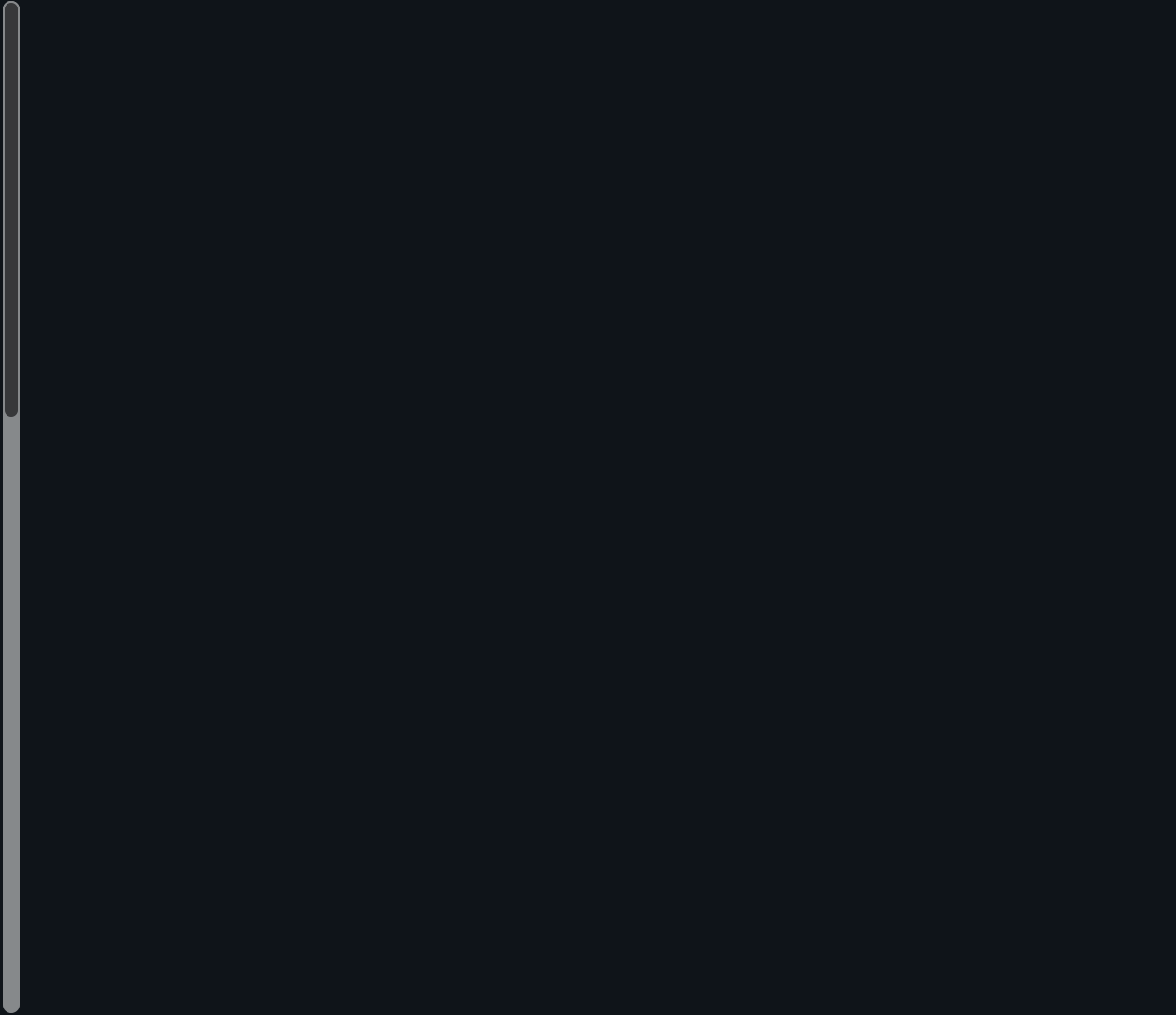
| Grade Letter | credit_card | debt_consolidation | educational | home_improvement | house | major_purchase | medical | moving | other | renewable_energy | small_business | vacation | wedding |
|--------------|-------------|--------------------|-------------|------------------|-------|----------------|---------|--------|-------|------------------|----------------|----------|---------|
| F            | 50.0%       | 51.5%              | 0.0%        | 37.0%            | 26.5% | 62.1%          | 54.2%   | 46.7%  | 34.6% | 25.0%            | 38.3%          | 25.0%    | 50.0%   |
| G            | 45.5%       | 46.7%              |             | 50.0%            | 28.6% | 33.3%          | 0.0%    | 0.0%   | 39.5% |                  | 42.1%          | 100.0%   |         |
| E            | 39.1%       | 41.0%              |             | 29.6%            | 20.0% | 45.1%          | 40.4%   | 37.0%  | 33.8% | 25.0%            | 38.9%          | 40.7%    | 16.7%   |
| D            | 29.2%       | 30.6%              |             | 27.3%            | 36.4% | 28.2%          | 30.8%   | 36.9%  | 25.4% | 14.3%            | 26.3%          | 18.2%    | 18.8%   |
| C            | 22.8%       | 22.6%              | 0.0%        | 19.0%            | 24.1% | 24.7%          | 25.7%   | 24.8%  | 19.1% | 15.4%            | 22.9%          | 17.1%    | 18.2%   |
| B            | 13.2%       | 13.4%              | 0.0%        | 15.0%            | 19.6% | 15.2%          | 15.3%   | 17.3%  | 15.0% | 16.7%            | 19.5%          | 9.7%     | 13.3%   |
| A            | 6.1%        | 6.2%               | 0.0%        | 5.0%             | 0.0%  | 5.6%           | 8.0%    | 6.1%   | 7.6%  | 40.0%            | 9.6%           | 15.3%    | 0.0%    |

# Loan Explorer — Drill into any segment

Expand grade → sub-grade → purpose for full segment-level detail

Segment detail table

| Grade Letter | Default Rate | Avg Int Rate | Avg DTI | Total Originated |
|--------------|--------------|--------------|---------|------------------|
| [-] A        | 6.2%         | 7            | 15.6    | \$143,602,600    |
| [+] 0        | 2.7%         | 6            | 14.0    | \$27,706,225     |
| [+] 1        | 4.6%         | 7            | 15.2    | \$21,385,975     |
| [+] 2        | 5.6%         | 7            | 15.8    | \$21,830,900     |
| [+] 3        | 7.2%         | 8            | 15.8    | \$32,572,675     |
| [+] 4        | 8.9%         | 8            | 16.6    | \$40,106,825     |
| [-] B        | 13.6%        | 11           | 17.4    | \$230,607,150    |
| [+] 5        | 10.7%        | 9            | 16.6    | \$42,669,800     |
| [+] 6        | 11.8%        | 10           | 17.1    | \$43,414,975     |
| [+] 7        | 13.4%        | 11           | 17.6    | \$48,055,025     |
| [+] 8        | 14.5%        | 11           | 17.8    | \$48,880,800     |
| [+] 9        | 17.1%        | 12           | 17.9    | \$47,586,550     |
| [-] C        | 22.3%        | 14           | 19.0    | \$241,231,325    |
| [+] 10       | 18.6%        | 13           | 18.3    | \$51,650,150     |
| [+] 11       | 21.4%        | 13           | 18.7    | \$48,578,825     |
| [+] 12       | 22.0%        | 14           | 19.2    | \$48,402,450     |
| [+] 13       | 25.0%        | 15           | 19.1    | \$48,455,750     |
| [-] 14       | 25.1%        | 16           | 19.0    | \$48,114,150     |



# Methodology & Model Performance

0.73

Model AUC

16x

Risk Separation

## Approach

XGBoost classifier trained on 500K stratified loans (2007–2018), 47 engineered features, 23 leakage fields excluded. Isotonic calibration applied. Random train/test split.

## Performance

AUC 0.73 on held-out test set. Top decile captures ~37% of all defaults despite holding only 10% of loans. Risk separation: 16× between deciles 1 and 10.

## Limitations

- Random split (not forward-time) — may overstate generalization to future vintages
- LGD assumed flat at 75%
- Lending Club's grade pricing already covers losses at portfolio level; model adds value at sub-segment level, not headline default prediction
- Test window 2007–2018 — pre-COVID, pre-rate-hike era

## Calibration: predicted vs actual default by decile

